When Target Choice Changes Benchmark Conclusions in Transformed Conversational Memory

Sugam Panthi
University of Southern Mississippi
sugam.panthi@usm.edu

Abstract

Conversational-memory benchmarks usually assume that each query has a single correct retrieval target. In transformed-memory stores, however, the same source evidence may appear as a raw dialogue turn, a derived memory, or a canonical fact, so benchmark conclusions can change even when ranked outputs stay fixed and only the scored target changes. We study this target non-invariance on LoCoMo and LongMemEval-S by rescoring saved retrieval traces under Raw, Source, and Canonical targets using TIAP, a fixed-output audit implemented in MTEL-Mem. Across four native retrievers and four external transfer settings, raw-versus-canonical nDCG changes on 83.4%–93.1% of shared queries, transfer results split between monotonic (Source > Canonical > Raw) and non-monotonic (Source > Raw > Canonical) orderings, and a fixed density sweep yields different winning configurations under different scored targets. These results show that, in the setting studied here, target choice is itself an evaluation decision that benchmarks should report explicitly.

1 Introduction

Conversational-memory benchmarks are increasingly used to justify memory-system choices (Maharana et al., 2024; Yu et al., 2024; Tavakoli et al., 2026; Li et al., 2024; Ong et al., 2024). That practice assumes the benchmark is rewarding the right retrieved item. In transformed-memory stores, that assumption can fail. The same source evidence may appear as the original dialogue turn, a derived memory, or a canonical fact prepared for serving.

Figure 1 shows the failure mode. The retrieved item at rank 3 is unchanged. What changes is only which stored form is allowed to count as correct. Under a raw-turn target, the retrieval is a miss; under source-family and canonical targets, it is a hit. If the scored target changes the measured winner, then the benchmark can recommend the wrong retriever, store density, or serving representation for the system being evaluated.

Classical IR has long studied evaluation sensitivity under assessor disagreement, alternative gains, and different retrieval units (Bailey et al., 2008; Yang et al., 2019; Craswell et al., 2020; Voorhees, 2001; Voorhees and Harman, 2005; Manning et al., 2008). Our setting is close in spirit but differs in one important way: prior sensitivity analyses usually vary judgments over a fixed unit space, whereas here the set of stored IDs that can receive credit may itself depend on how the memory store is built.

We study this source-linked transformed-memory setting on LoCoMo and LongMemEval-S. On shared queries alone, switching only the scoring target changes nDCG for 83.4% to 93.1% of Raw-versus-Canonical comparisons across our two benchmarks. Transfer settings split between monotonic and non-monotonic target orderings, and on a fixed density sweep the winning store configuration also changes with the scored target. These results show the benchmark-validity failure mode directly on fixed ranked outputs.

We measure the effect with TIAP, a fixed-output audit based on rescoring saved traces from earlier runs under three target sets and repeating the main comparisons. To make that audit reusable, we implement it in MTEL-Mem, a reproducible evaluation layer over existing benchmarks.

Figure 1: Stylized example of target non-invariance. The retrieved item at rank 3 is unchanged; only the scoring target differs.

Query	"When did the user move to Austin?"
Raw turn (turn #47)	"Yeah I actually just relocated to Austin last summer, it's been great so far."
Observation	"User relocated to Austin in summer 2021."
Canonical fact	"User lives in Austin (since 2021)."
Retrieved at rank 3	"User lives in Austin (since 2021)."
Raw-turn target	MISS: rank-3 item is not turn #47's raw ID.
Source target	HIT: rank-3 item derives from turn #47.
Canonical target	HIT: rank-3 item is a canonical memory for turn #47.

TIAP is a sensitivity audit, not a verdict on which target is correct. Whether a permissive target deserves credit depends on the system being evaluated and should be checked, not assumed. MTEL-Mem makes that audit reproducible on existing benchmarks, and the rest of the paper measures how often target choice changes scores, system orderings, and practical design recommendations.

2 Related Work

Conversational-memory benchmarks such as LoCoMo, LongMemEval, and BEAM have established retrieval- and QA-based evaluation for long-horizon dialogue systems (Maharana et al., 2024; Yu et al., 2024; Tavakoli et al., 2026). Memory-system papers such as Mem0, Theanine, RMM, MemWalker, and ReadAgent focus instead on architecture, storage policy, and memory management (Chhikara et al., 2025; Ong et al., 2024; Tan et al., 2025; Chen et al., 2023; Lee et al., 2024). We ask a narrower evaluation question: when a system stores raw turns, derived memories, and canonical facts linked to the same source evidence, which of those stored IDs should the benchmark be allowed to credit?

The closest analogy is classical IR work on evaluation sensitivity (Bailey et al., 2008; Yang et al., 2019; Craswell et al., 2020; Voorhees, 2001; Voorhees and Harman, 2005; Manning et al., 2008; Järvelin and Kekäläinen, 2002). Document-versus-passage evaluation is especially relevant because the underlying evidence may be unchanged while the scored unit changes (Yang et al., 2019; Craswell et al., 2020). BEIR likewise shows that retrieval conclusions depend on benchmark design choices even when the judged unit is fixed in advance (Thakur et al., 2021). The key difference in our setting is that the admissible target set can depend on store construction: prior sensitivity work varies judgments over fixed units, whereas we vary which source-linked stored descendants are eligible to receive credit in the first place.

RAG evaluation frameworks such as RAGAs and ARES are also adjacent because they evaluate whether retrieved evidence supports an answer (Es et al., 2024; Saad-Falcon et al., 2024). What they usually hold fixed, however, is the candidate evidence unit itself. Our audit addresses the earlier question that transformed-memory benchmarks leave open: before answer-level evaluation begins, which stored descendants of the same source evidence should count as admissible retrieval targets?

3 Problem Setting and Audit Protocol

Let q be a benchmark query, let S denote a memory store, let $R_k^S(q)$ be the top-k ranked output emitted by S, and let $s(q)$ denote the source evidence that anchors the gold label. We use *scoring target* for the set of stored IDs that can receive credit. In the setting we study, Raw credits only the exact source-turn memory, Source credits any stored descendant linked to the same source evidence, and Canonical credits transformed non-raw descendants intended for serving. Formally, for a fixed source anchor $s(q)$ and store S, we define:

$$T_{\text{raw}}(q, S) = \{\text{raw-turn IDs}\}, \tag{1}$$

$$T_{\text{src}}(q, S) = \{\text{all source-linked descendant IDs}\}, \tag{2}$$

$$T_{\text{can}}(q, S) = \{\text{canonical non-raw IDs}\}. \tag{3}$$

By construction, $T_{\text{raw}}(q, S) \subseteq T_{\text{src}}(q, S)$ and $T_{\text{can}}(q, S) \subseteq T_{\text{src}}(q, S)$, while $T_{\text{raw}}(q, S)$ and $T_{\text{can}}(q, S)$ are usually disjoint. When the lineage contract is shared across compared runs, we omit S for readability.

The audit question is simple: what changes when only the target set changes? TIAP answers that question by holding ranked outputs fixed, rescoring each saved run under every target set, repeating comparisons on shared query subsets when coverage differs, and then inspecting query-level disagreement plus a sampled semantic audit of contested credit. Algorithm 1 summarizes the procedure.

All main results come from fixed-output rescoring. We keep each system's ranked traces un-

changed and only change which target set is allowed to receive credit. The reported metric shifts, matched-subset deltas, and winner flips are therefore properties of the scoring target under the released traces and mappings, not of rerunning retrieval under different settings. A reproducer can verify the core result directly from the saved traces, target mappings, and scorer.

For systems A and B, metric M, and target t, define

$$\Delta_t(A, B) = M_t(A) - M_t(B). \tag{4}$$

If the sign of $\Delta_t(A, B)$ changes when only t changes, then the benchmark conclusion is not target-invariant. This lets us test whether the benchmark's existing conclusion survives target-set changes under fixed ranked outputs.

This differs from standard assessor-disagreement settings in one important way. In those settings, the items that can receive credit are fixed and only the judgments change. Here, the set of stored IDs that can receive credit can change with the store because different systems may expose different descendants of the same source evidence. If one system retrieves the raw turn and another retrieves a canonical descendant that preserves the same source evidence, then a benchmark that credits only the raw ID has already fixed the answer before scoring begins.

4 Experimental Setup

4.1 Benchmarks and Target Mappings

We instantiate the audit on LoCoMo and LongMemEval-S using four retrievers: lexical search, all-MiniLM-L6-v2, BGE-M3, and mxbai-embed-large. All results are computed from saved top-$k = 60$ ranked traces and rescored under the source-linked target sets used in this paper. For each query, the source evidence is mapped to Raw, Source, and Canonical targets through source-linked lineage metadata.

The primary native store contains the raw turn plus parser-derived transformed memories linked back to the same source evidence. Source unions raw and transformed IDs for a source fixture. Canonical keeps only non-raw transformed IDs. Canonical coverage therefore differs from Raw coverage, which is why shared-subset rescoring is part of the main analysis rather than an optional follow-up. For the density sweep, F_1, F_5, and F_8 denote *parser_max_facts* $= 1, 5, 8$ transformed facts per source turn.

4.2 Retrieval Setup and Inferential Scope

The four native retrievers are lexical search, all-MiniLM-L6-v2, BGE-M3, and mxbai-embed-large. All native results use top-$k = 60$, extractive answer mode, and random seed `1337`. The lexical run uses the same benchmark harness without dense embeddings, and embedding providers are served locally via Ollama.

Shared-subset paired statistics are computed from fixed ranked traces with paired bootstrap resampling. The within-provider summary uses 3,000 paired resamples, percentile confidence intervals, and RNG seed `1337`. Because all reported differences come from rescoring saved outputs, they isolate evaluation-target changes rather than retrieval changes. Reproducing the main audit requires the benchmark queries, source-linked target mappings, saved ranked traces, and the rescoring scripts; a full end-to-end rerun is useful but not necessary to verify the central fixed-output result.

The transfer analysis includes full LoCoMo and LongMemEval-S runs for both Mem0 and MemoryOS. Zep is out of scope because its graph-native memories do not expose a stable source-turn-descendant ID mapping comparable to the Raw/Source/Canonical sets used here.

4.3 Reproducibility Layer

To make the audit reusable, we implement it in MTEL-Mem, an evaluation layer over existing conversational-memory benchmarks. The layer takes saved ranked traces as input and reproduces the same rescoring, matched-subset comparisons, instability summaries, and semantic-audit artifacts used in this paper.

MTEL-Mem currently wraps LoCoMo and LongMemEval-S. For each benchmark, it packages adapters, target mappings, source fixtures, category metadata, and reporting utilities into one reproducible layer. The input contract is intentionally simple: a new system exports ranked traces with benchmark name, query ID, retrieved memory ID, rank, and optional score. Given those traces and the benchmark adapter, the layer can reproduce the audit without requiring a new benchmark release.

4.4 Semantic-Audit Methodology

The semantic audit covers 120 `raw_miss / source_hit / canonical_hit` disagree-

Table 1: Algorithm 1: TIAP as an implementable audit procedure.

Step	Inputs	Operation	Outputs
0	Query set Q; source anchor $s(q)$; saved ranked traces $R_k^S(q)$; admissible ontology family $T_t(q, S)$; evaluation metrics M	Initialize the audit on fixed ranked outputs only. No retrieval reruns are allowed inside TIAP.	Auditable run bundle $\{Q, s(q), R_k^S(q), T_t(q, S), M\}$.
1	Q, $s(q)$, lineage contract for each system or configuration S	Enumerate the admissible raw, source-family, and canonical descendant IDs for every query. If compared systems expose different descendant families, keep the benchmark source fixed and record ontologies separately for each S.	Per-system ontology map $T_t(q, S)$.
2	$R_k^S(q)$, $T_t(q, S)$, M	Rescore each saved run under every admissible ontology t to obtain $M_t(S)$.	Per-target metrics for each run or system.
3	$M_t(S)$, ontology coverage by query	If ontology coverage differs, restrict all ontologies to the same shared query subset before comparing systems or configurations. Compute paired deltas $\Delta_t(A, B) = M_t(A) - M_t(B)$ on the same queries for every t.	Shared-subset metrics, paired deltas, and optional confidence intervals.
4	Per-query rescored outputs from Step 2	Count how often target switching changes hit status, top-1 correctness, graded utility, or the winning system / configuration on the same fixed outputs.	Query-level instability summaries and winner-flip counts.
5	Disagreement cases from Step 4 where provenance-relaxed targets award extra credit	Sample disagreement rows and label whether the credited transformed memory actually supports the query. Report audit scope and reliability separately from aggregate benchmark metrics.	Semantic-audit summary over contested credit.
6	Steps 2–5	Report per-target metrics, shared-subset deltas, instability summaries, and any semantic-audit caveat. Flag target non-invariance when the sign of $\Delta_t(A, B)$ changes, when a winner flips across admissible targets, or when the semantic audit shows that a permissive ontology is not reliable enough to stand alone.	Final audit conclusion and reporting recommendation.

Table 2: Dataset coverage used by the audit. Native query counts vary slightly by provider because some runs cover slightly different benchmark subsets; BEAM-1M is a supplementary lexical-only stress test rather than part of the eight-run native matrix.

Dataset	Fixture	Eval	Native runs	Canon. cov.
LoCoMo	5,882	1,977	1,510–1,533	899–963
LongMemEval-S	231,595	470	439–470	299–332
BEAM-1M (supp.)	74,630	625	625	408

ment cases. We selected 60 LongMemEval-S all-MiniLM cases and 60 LoCoMo BGE-M3 cases deterministically, stratified by credited-rank bucket (1–5, 6–20, 21–60) and then evenly spaced by query index within each bucket. Labels follow a three-way rubric: `supports`, `partial`, and `does_not_support`. Auxiliary serving-semantics tags are reported only as exploratory notes and are not used as primary evidence.

The main audit sheet was labeled internally, and a second human reviewer independently labeled a deterministic 36-case overlap subset. Because the label space is ordinal and most disagreements are adjacent boundary calls around `partial`, we use weighted agreement as the main reliability summary: linear-weighted $\kappa = 0.618$ and quadratic-weighted $\kappa = 0.755$. We then adjudicated the 14 disagreement rows; only two changed relative to the internal labels.

5 Main Empirical Findings

5.1 Subset-Controlled Headline Result

The main result comes from rescoring the same saved runs on the same matched query subsets. Target coverage differs across target sets, so the full-run overview is less informative. Switching only the scoring target changes nDCG on 83.4%–93.1% of shared Raw-versus-Canonical queries across our two benchmarks. On LongMemEval-S, Canonical exceeds Raw for every provider on the shared subset, with the largest Canonical–Raw effects on BGE-M3 (+0.336 [0.302, 0.371]) and mxbai-embed-large (+0.420 [0.382, 0.457]). On LoCoMo, Canonical also exceeds Raw for every provider, but Source remains above Canonical for

all four providers, with the strongest reversal on BGE-M3 (-0.088 Canonical–Source, 95% CI excluding zero). This means target choice changes both effect size and which target looks most favorable, even when the underlying ranked outputs are fixed.

The same instability appears at the query level. Across shared queries, Raw-versus-Canonical rescoring changes nDCG on 83.6%–93.2% of major LoCoMo categories and 77.1%–100.0% of LongMemEval-S categories. The appendix reports the full query-level matrix.

5.2 Store-Design Recommendation Flips

The next question is whether target sensitivity changes a practical engineering recommendation. In a parser-density sweep on a fixed matched subset ($n = 453$), the answer is yes. Here F_1, F_5, and F_8 denote *parser_max_facts* $= 1, 5, 8$. Changing only the scored target changes which configuration appears best.

For example, in the lexical F_1 versus F_5 comparison, Raw prefers the sparser F_1 configuration, Source prefers the denser F_5 configuration, and Canonical returns to F_1. Across the six pairwise comparisons retained in Table 10, Source prefers the denser setting in all six comparisons, Raw prefers the sparsest setting in four of six, and Canonical is split. Target choice therefore changes the recommended store design even though the compared systems and saved outputs are otherwise fixed.

5.3 Query- and Category-Level Instability

The effect is not confined to one query family. On LoCoMo, Canonical coverage is highest for Multi-hop queries (84.7–88.3%) and lower for Single-hop and Temporal queries (50.9–60.9%), yet Raw-versus-Canonical nDCG still changes on 83.6–93.2% of shared Single-hop, Temporal, and Multi-hop queries. Open-domain is lower but still changes on 61.7–71.9% of shared queries.

LongMemEval-S shows the same broad pattern. Canonical coverage is highest for multi-session and knowledge-update queries (75.0–86.5%) and lower for single-session-user and single-session-preference queries (52.4–56.7%), but Raw-versus-Canonical nDCG still changes on 77.1%–100.0% of shared queries across every listed category. One nuance is that Source-versus-Canonical instability on LongMemEval-S is concentrated in multi-session and knowledge-update questions and is close to zero for single-session-assistant questions. That suggests provenance relaxation matters less when the answer is already concentrated in a simple one-session assistant fact. Appendix E shows the same distinction concretely: for the asylum-application query in Table 11, Source receives credit because the retrieved memory comes from the correct source turn, while Canonical withholds credit because the retrieved transformed item is not the canonical representative.

6 Additional Validation and Caveats

6.1 Semantic Audit of Contested Credit

A natural concern is that transformed-memory gains may reflect over-permissive credit rather than semantically correct retrieval. We therefore audited 120 disagreement cases in which transformed memories receive credit where Raw does not: 60 LongMemEval-S all-MiniLM cases and 60 LoCoMo BGE-M3 cases drawn from `raw_miss / source_hit / canonical_hit` disagreements.

The audit shows two things at once. Some target-conditioned gains recover semantically meaningful evidence that Raw misses. But a nontrivial fraction of contested gains are also over-permissive. Across the 120 audited cases, 28 fully support the query, 50 are partially supportive, and 42 do not support it. Among the non-supportive cases, the most common failure modes are missing a required detail such as a date or entity (27/42), retrieving generic context rather than answer-bearing evidence (8/42), and retrieving the wrong event or timeline (7/42).

The semantic audit therefore shows that target choice changes what the benchmark counts as success and needs to be checked explicitly, not assumed to define a universally correct relaxed target. The overlap review supports this reading without overclaiming certainty: linear-weighted $\kappa = 0.618$ overall, quadratic-weighted $\kappa = 0.755$, and only two label changes after adjudication. At the same time, the 120-case sample is too small for strong prevalence claims, so we treat it as a scoped caveat analysis rather than a final estimate of semantic validity.

6.2 Cross-Architecture Transfer

To test whether the effect extends beyond the primary parser-derived store, we also evaluate Mem0 and MemoryOS. The transfer runs remain strongly target-sensitive, but the ordering is not uniform. On

Table 3: Fixed-subset rescoring on canonical-covered shared subsets for all eight native runs. Canonical exceeds Raw on every shared subset; Source exceeds Canonical on all four LoCoMo shared subsets.

Dataset	Provider	n	Raw	Source	Canonical	Canonical–Raw	Canonical–Source
LoCoMo	Lexical	899	0.1876	0.3809	0.3556	+0.168 [0.147, 0.189]	-0.025 [-0.036, -0.015]
	all-MiniLM	899	0.2118	0.4198	0.3875	+0.176 [0.154, 0.198]	-0.032 [-0.043, -0.022]
	BGE-M3	963	0.2548	0.4109	0.3231	+0.068 [0.045, 0.091]	-0.088 [-0.101, -0.075]
	mxbai-embed-large	899	0.2120	0.4195	0.3870	+0.175 [0.152, 0.198]	-0.033 [-0.043, -0.022]
LongMemEval-S	Lexical	332	0.1877	0.4041	0.4558	+0.268 [0.236, 0.303]	+0.052 [0.038, 0.066]
	all-MiniLM	332	0.2001	0.4561	0.5309	+0.331 [0.299, 0.364]	+0.075 [0.061, 0.089]
	BGE-M3	299	0.1969	0.4576	0.5326	+0.336 [0.302, 0.371]	+0.075 [0.061, 0.090]
	mxbai-embed-large	332	0.1815	0.5032	0.6015	+0.420 [0.382, 0.457]	+0.098 [0.083, 0.113]

Table 4: Completed 120-query semantic audit of `raw_miss` / `source_hit` / `canonical_hit` disagreements.

Dataset	Supports	Partial	Does not support
LongMemEval-S	10	24	26
LoCoMo	18	26	16
Total	28	50	42

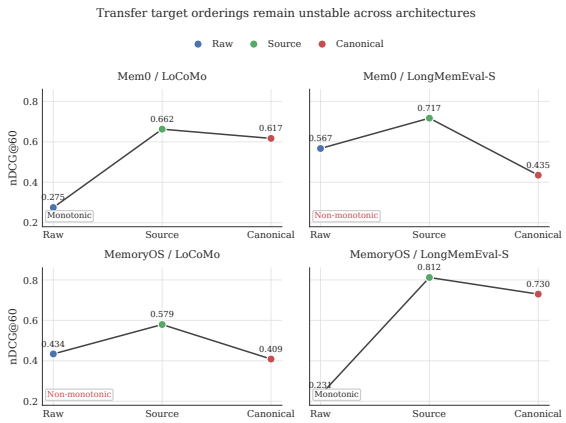

Figure 2: Small-multiples view of the transfer summary. Two settings are monotonic (Source > Canonical > Raw) and two are non-monotonic (Source > Raw > Canonical), so transfer broadening preserves target sensitivity but not a single abstraction direction.

full LoCoMo runs, Mem0 is monotone (Source > Canonical > Raw), while MemoryOS shows the non-monotonic pattern Source > Raw > Canonical. On full LongMemEval-S runs, MemoryOS is monotone whereas Mem0 is non-monotonic.

The target effect is therefore not confined to one benchmark or one native store design. It is also not just rewarding a larger target set, because different architectures produce different target orderings. We still treat this as breadth rather than full generalization: two additional architectures are evidence against a one-off artifact, not proof of invariance across arbitrary memory pipelines.

Table 5: Architectural transfer summary (full-run nDCG@60).

System / Data	Raw	Source	Canonical
Mem0 / LoCoMo	0.2749	0.6625	0.6169
Mem0 / LongMemEval-S	0.5670	0.7170	0.4350
MemoryOS / LoCoMo	0.4339	0.5794	0.4086
MemoryOS / LongMemEval-S	0.2306	0.8121	0.7298

Table 6: Supplementary BEAM-1M lexical stress test. The ordering remains Raw < Source < Canonical both on the full run and on the canonical-covered shared subset.

Slice	Q	Raw	Source	Canonical
Full run	625	0.0554	0.0969	0.1969
Shared subset	408	0.0706	0.1342	0.1969

6.3 Supplementary BEAM Stress Test

We also include a supplementary lexical stress test on BEAM's grounded 1M slice (Tavakoli et al., 2026). This result is deliberately secondary: one provider, one benchmark scale, and no attempt to fold BEAM into the main native matrix. Even under that restrained setup, the target-sensitive pattern remains clear. On the full 625-query run, nDCG@60 moves from 0.055 to 0.097 to 0.197. On the canonical-covered shared subset ($n = 408$), the same ordering persists with $0.071 < 0.134 < 0.197$.

We interpret this as a supplementary stress test, not a third headline benchmark. The appendix includes supporting 100K lexical and mxbai runs that show the same ordering.

An auxiliary answer-level alignment check is mixed across datasets and is reported in Appendix F.

7 Practical Reporting Guidance

Taken together, the results suggest a conservative reporting practice for transformed-memory evaluation. The goal is to make target sensitivity visi-

ble whenever store construction creates multiple source-linked targets and to give readers a rough sense of when each target is more trustworthy.

Our category-level results suggest one useful boundary. When the query is a simple single-session fact, target instability is lower. On LongMemEval-S, Source-versus-Canonical instability is close to zero for single-session-assistant questions, which makes Raw a reasonable safety floor and often makes Canonical easier to justify when the deployed system actually serves transformed memories. When the query is multi-session, knowledge-updating, or temporally sensitive, instability is much higher. Those are the cases where target choice most often changes the conclusion, so they are the strongest cases for running TIAP and reporting more than one target.

The auxiliary answer-alignment check in Appendix F suggests a second calibration signal. On LoCoMo, Raw aligns better with answer F1, while on LongMemEval-S, Source and Canonical align better. That means target reporting should follow both serving semantics and benchmark behavior: if transformed targets align better with answer quality on a benchmark, Canonical can be the primary report with Raw as a floor; if alignment is mixed, report multiple targets rather than forcing one default.

Source remains best treated as a stress test unless semantic auditing shows that provenance-linked descendants are reliably sufficient for the queried fact. Across all settings, shared-subset rescoring should be used whenever coverage differs, and winner flips should be reported explicitly when store density, retriever choice, or system ordering changes with the target.

8 Conclusion

The problem studied in the paper is narrow, but the implications for how the field reports retrieval results are not. When a memory store exposes multiple stored forms of the same evidence, a benchmark that silently credits only one of them has already made a design choice before scoring begins. Which can reverse which system wins, which store density is recommended, and whether a retrieval gain looks real or not. The practical takeaway from our results is that simple single-session queries are the closest to a stable case: target choice rarely reverses conclusions there. It is the multi-session, knowledge-updating, and temporally sensitive queries, the ones that arguably matter most for deployed memory systems, where target sensitivity is highest. That asymmetry suggests that future benchmarks might distinguish these query types explicitly rather than reporting a single aggregate score. TIAP and MTEL-Mem are offered as infrastructure, not verdicts. The audit does not prescribe which target is correct; it makes the cost of that choice visible. Whether the field converges on a reporting convention, a richer target ontology, or per-query target disclosure is an open question. We hope future work on conversational memory evaluation treats the scored target as a design choice to be reported, not a default to be inherited.

9 Limitations

This study is limited to transformed conversational memory rather than IR evaluation in general. The native headline matrix is still concentrated in one parser-derived store construction. The transfer results broaden the evidence to Mem0 and MemoryOS, but that remains a small architectural sample.

The semantic audit covers 120 disagreement cases rather than a full blind annotation campaign across all queries and target pairs. The overlap review is also modest. It is enough to characterize likely failure modes, but not enough to settle prevalence precisely.

TIAP currently treats target membership as binary once a retrieved item falls inside a target set. It does not assign graded credit to partially sufficient evidence or compose support across multiple retrieved facts. That matters most for multi-hop and multi-session questions.

The Raw / Source / Canonical basis used here is a minimal audit basis for the source-linked transformed-memory systems we study. It is useful because these systems expose recoverable lineage from transformed memories back to benchmark source evidence. Other memory architectures may expose different stored units and may require different target sets or audit constructions. We therefore treat this basis as an instantiation for this setting, not as a complete taxonomy of conversational memory. The audit also assumes stable source anchors $s(q)$, recoverable lineage from retrieved IDs back to those anchors, enumerable target sets over the descendants, and exported fixed ranked traces. Parser- or extraction-based stores satisfy this directly, which is why our current instantiation covers the native store, Mem0, and Memo-

ryOS. Graph-native memories may require node-, edge-, or path-level target definitions beyond the Raw/Source/Canonical basis used here.

Our auxiliary answer-alignment analysis is also mixed rather than decisive. Raw aligns better on LoCoMo, while Source and Canonical align better on LongMemEval-S. For that reason, we do not claim that one target ontology best predicts downstream answer quality across settings.

More broadly, TIAP and MTEL-Mem are meant to make target sensitivity auditable, not to prescribe a universal ontology or a single pass/fail threshold. Extending the audit to graph-native memories and other retrieval settings remains future work.

10 Availability Statement

An anonymized implementation of MTEL-Mem, including target mappings, rescoring code, example ranked traces, and semantic-audit materials sufficient to reproduce the main analyses in this paper, is included in the supplementary materials.

11 Ethics Statement

This paper uses public benchmark data and stored system traces. The primary ethical issue is evaluation transparency: reporting only one admissible target ontology can overstate or understate system quality, so the protocol is intended to reduce misleading benchmark claims. That matters downstream because benchmark-guided memory design choices can affect which information deployed systems retain, surface, or suppress for users.

References

Peter Bailey, Nick Craswell, Ian Soboroff, Paul Thomas, Arjen P. de Vries, and Emine Yilmaz. 2008. Relevance assessment: Are judges exchangeable and does it matter? In *Proceedings of the 31st Annual International ACM SIGIR Conference on Research and Development in Information Retrieval*, pages 667–674.

Howard Chen, Ramakanth Pasunuru, Jason Weston, and Asli Celikyilmaz. 2023. Walking down the memory maze: Beyond context limit through interactive reading. *arXiv preprint arXiv:2310.05029*.

Prateek Chhikara, Dev Khant, Saket Aryan, Taranjeet Singh, and Deshraj Yadav. 2025. Mem0: Building production-ready AI agents with scalable long-term memory.

Nick Craswell, Bhaskar Mitra, Emine Yilmaz, Daniel Campos, and Ellen M. Voorhees. 2020. Overview of the TREC 2019 deep learning track. *arXiv preprint arXiv:2003.07820*.

Shahul Es, Jithin James, Luis Espinosa Anke, and Steven Schockaert. 2024. RAGAs: Automated evaluation of retrieval augmented generation. In *Proceedings of the 18th Conference of the European Chapter of the Association for Computational Linguistics: System Demonstrations*, pages 150–158.

Kalervo Järvelin and Jaana Käkäläinen. 2002. Cumulated gain-based evaluation of IR techniques. *ACM Transactions on Information Systems*, 20(4):422–446.

Kuang-Huei Lee, Xinyun Chen, Hiroki Furuta, John Canny, and Ian Fischer. 2024. A human-inspired reading agent with gist memory of very long contexts. *arXiv preprint arXiv:2402.09727*.

Hao Li, Chenghao Yang, An Zhang, Yang Deng, Xiang Wang, and Tat-Seng Chua. 2024. Hello again! LLM-powered personalized agent for long-term dialogue.

Adyasha Maharana, Dong-Ho Lee, Sergey Tulyakov, Mohit Bansal, Francesco Barbieri, and Yuwei Fang. 2024. Evaluating very long-term conversational memory of LLM agents. In *Proceedings of the 62nd Annual Meeting of the Association for Computational Linguistics (Volume 1: Long Papers)*, pages 13851–13870.

Christopher D. Manning, Prabhakar Raghavan, and Hinrich Schütze. 2008. *Introduction to Information Retrieval*. Cambridge University Press.

Kai Tzu-iunn Ong, Namyoung Kim, Minju Gwak, Hyungjoo Chae, Taeyoon Kwon, Yohan Jo, Seungwon Hwang, Dongha Lee, and Alice Oh. 2024. Towards lifelong dialogue agents via timeline-based memory management.

Jon Saad-Falcon, Omar Khattab, Christopher Potts, and Matei Zaharia. 2024. ARES: An automated evaluation framework for retrieval-augmented generation systems. In *Proceedings of the 2024 Conference of the North American Chapter of the Association for Computational Linguistics: Human Language Technologies (Volume 1: Long Papers)*, pages 338–354.

Zhen Tan, Jun Yan, I-Hung Hsu, Rujun Han, Zifeng Wang, Long T. Le, Yiwen Song, Yanfei Chen, Zhaolin Xu, and Diyi Yang. 2025. In prospect and retrospect: Reflective memory management for long-term personalized dialogue agents.

Mohammad Tavakoli, Alireza Salemi, Carrie Ye, Mohamed Abdalla, Hamed Zamani, and J Ross Mitchell. 2026. Beyond a million tokens: Benchmarking and enhancing long-term memory in llms. *arXiv preprint arXiv:2510.27246*.

Nandan Thakur, Nils Reimers, Andreas Rücklé, Abhishek Srivastava, and Iryna Gurevych. 2021. BEIR: A heterogeneous benchmark for zero-shot evaluation of information retrieval models. In *Proceedings of*

the Neural Information Processing Systems Track on Datasets and Benchmarks.

Ellen M. Voorhees. 2001. The philosophy of information retrieval evaluation. In *Cross-Language Evaluation Forum: CLEF 2001*.

Ellen M. Voorhees and Donna K. Harman, editors. 2005. *TREC: Experiment and Evaluation in Information Retrieval*. MIT Press.

Wei Yang, Kuang Lu, Peilin Yang, and Jimmy Lin. 2019. Critically examining the "neural hype": Weak baselines and the additivity of effectiveness gains from neural ranking models. In *Proceedings of the 42nd International ACM SIGIR Conference on Research and Development in Information Retrieval*, pages 1129–1132.

Zichen Yu, Jiale Li, Yifei Wang, Weiran Yang, Yafeng Ding, Yue Zhang, Yansong Wang, Weizhi Hu, Dong Ma, Zhicheng Liu, Haobo Wang, Hsuan-Ting Chen, Kaijie Zhang, Xiang Wang, Ming Zhen, Aixi Tao, Rajarshi Mitra, Jing Weng, Yuxi Yao, Xiang Yue, Qian Zhao, Huan Wang, Fei Huang, Yong Wang, Xin Wang, Yizhu Zhang, Shibo Hao, Lu Wang, Yao Lu, Chao Liu, Yu Liu, Da Yin, and Daxin Jiang. 2024. Benchmarking chat assistants on long-term interactive memory. *arXiv preprint arXiv:2410.10813*.

A Query-Level Instability

Table 7 reports the full query-level disagreement matrix across all eight provider–dataset pairs.

Table 7: Query-level instability between target definitions across all eight main-matrix provider-dataset pairs.

Dataset	Provider	Pair	Shared queries	Hit flips	Top-1 flips	nDCG changed	Change rate
LoCoMo	Lexical	Raw vs Source	1,533	181	173	526	34.3%
		Raw vs Canonical	899	244	193	758	84.3%
		Source vs Canonical	899	63	20	397	44.2%
	all-MiniLM	Raw vs Source	1,533	162	198	541	35.3%
		Raw vs Canonical	899	226	217	794	88.3%
		Source vs Canonical	899	64	19	425	47.3%
	BGE-M3	Raw vs Source	1,510	134	167	473	31.3%
		Raw vs Canonical	963	259	214	851	88.4%
		Source vs Canonical	963	125	47	532	55.2%
	mxbai-embed-large	Raw vs Source	1,533	162	190	541	35.3%
		Raw vs Canonical	899	226	208	794	88.3%
		Source vs Canonical	899	64	18	425	47.3%
LongMemEval-S	Lexical	Raw vs Source	470	50	113	259	55.1%
		Raw vs Canonical	332	68	113	277	83.4%
		Source vs Canonical	332	18	0	158	47.6%
	all-MiniLM	Raw vs Source	470	70	129	301	64.0%
		Raw vs Canonical	332	78	129	309	93.1%
		Source vs Canonical	332	8	0	158	47.6%
	BGE-M3	Raw vs Source	439	62	112	276	62.9%
		Raw vs Canonical	299	67	112	281	94.0%
		Source vs Canonical	299	5	0	144	48.2%
	mxbai-embed-large	Raw vs Source	470	130	152	311	66.2%
		Raw vs Canonical	332	131	152	312	94.0%
		Source vs Canonical	332	1	0	154	46.4%

B External Transfer Detail

Table 8 provides the detailed transfer results for the external architectures.

Table 8: External transfer validation on Mem0 and MemoryOS across LoCoMo and LongMemEval-S.

System / Data	Target	Q	R@60	MRR	nDCG@60
Mem0 / LoCoMo	Raw turn	1977	0.723	0.158	0.275
	Source	1977	0.945	0.620	0.663
	Canonical	1977	0.901	0.577	0.617
Mem0 / LongMemEval-S	Raw turn	470	0.978	0.454	0.567
	Source	470	0.987	0.687	0.717
	Canonical	470	0.758	0.412	0.435
MemoryOS / LoCoMo	Raw turn	1977	0.770	0.364	0.434
	Source	1977	0.994	0.481	0.579
	Canonical	1977	0.990	0.265	0.409
MemoryOS / LongMemEval-S	Raw turn	470	0.359	0.207	0.231
	Source	470	0.995	0.789	0.812
	Canonical	470	0.989	0.682	0.730

C Supplementary BEAM Runs

Table 9 summarizes the supporting BEAM-100K runs kept out of the main text. We use them as appendix evidence that the BEAM effect is already visible at a smaller scale and under a second embedder, while keeping the million-token lexical run as the single main-text BEAM citation.

D Decision-Level Density Example

Table 10 reports the sign-only density sweep referenced in the main text.

Table 9: Supporting BEAM-100K runs. Both the lexical and mxbai settings preserve the same Raw < Source < Canonical ordering on the full run and on the canonical-covered shared subset.

Scale	Setting	Q	Raw	Source	Canonical
100K	Lexical full	355	0.1105	0.2064	0.3489
100K	Lexical shared subset	246	0.1430	0.2815	0.3489
100K	mxbai full	355	0.1233	0.2281	0.3898
100K	mxbai shared subset	245	0.1617	0.3135	0.3898

Table 10: Target-dependent winner by parser density (F_1, F_5, F_8) on the fixed shared subset ($n = 453$). The winning configuration depends on whether scoring uses Raw, Source, or Canonical.

Provider	**Comp.**	**Raw**	**Source**	**Canonical**
Lexical	F_1 vs F_5	F_1	F_5	F_1
	F_1 vs F_8	F_1	F_8	F_1
	F_5 vs F_8	F_5	F_8	F_8
all-MiniLM	F_1 vs F_5	F_1	F_5	F_1
	F_1 vs F_8	F_1	F_8	F_1
	F_5 vs F_8	F_8	F_8	F_8

E Qualitative Cases

Table 11 lists representative disagreement cases illustrating how target choice changes evaluation outcomes.

Table 11: Representative disagreement cases from the LongMemEval-S all-MiniLM-L6-v2 run.

Query	**Raw nDCG**	**Source nDCG**	**Canonical nDCG**	Interpretation
What degree did I graduate with?	0.000	0.613	1.000	A transformed memory at rank 1 recovers the correct fact; raw-turn scoring gives no credit because the verbatim source ID is absent.
How long is my daily commute to work?	0.173	1.000	1.000	The correct source-linked memory is retrieved immediately in transformed form, but the raw-turn match appears much later in the ranking.
What play did I attend at the local community theater?	0.000	0.500	0.500	A non-raw memory at rank 3 supports the answer, converting a raw miss into both a Source and Canonical hit.
How long did I wait for the decision on my asylum application?	—	0.171	0.000	Source credit is granted because a memory from the correct source turn is retrieved, but canonical scoring withholds credit because the retrieved transformed item is not the canonical representative.

F Auxiliary Answer-Level Alignment

We also ran an auxiliary query-level check that joins target-specific hit labels to answer F1 on the same saved traces. The result is mixed. On LoCoMo, Raw has the strongest average answer-alignment signal across the eight native runs. On LongMemEval-S, Source and Canonical align better, though on a thinner base of four native runs.

We treat this analysis as supplementary because it does not replace the paper's main result, which comes from fixed-output benchmark instability under different target sets. Instead, it provides an additional downstream check showing that answer-level alignment does not point to one universally preferred target set.

Table 12: Auxiliary query-level answer-alignment check on shared Raw-versus-Canonical eligible queries. 'Raw only' and 'Canonical only' denote disagreement cells where hit@60 changes under identical ranked outputs. Means are averaged over native full runs within each dataset; counts are summed across those runs.

Dataset	Raw-only n	Canon-only n	Raw-only F1	Canon-only F1	Raw-only strong	Canon-only strong
LoCoMo	507	1142	0.123	0.046	0.063	0.013
LongMemEval-S	32	312	0.030	0.065	0.000	0.004